Hidden orchard

The mathematics in one view

A nonzero lattice point T=(a,b) is **visible** from O=(0,0) when the open line segment OT contains no other lattice point. A blocker must lie exactly on that segment; O and T are not blockers. The packet uses the first quadrant, including the axes.

Exact criterion. For nonnegative integer coordinates, not both zero, let d=gcd(a,b). The first lattice point on the ray is (a/d,b/d), and the interior points are exactly k(a/d,b/d), for k=1,...,d-1. Thus T is visible if and only if d=1, and there are d-1 blockers. On an axis use gcd(a,0)=a; only the first unit point is visible.

What the geometry encodes. Every lattice ray has one primitive first point, with coprime coordinates. Its other lattice points are positive integer multiples of that point. Scaling a direction explains hidden dots; a common divisor measures how many equal primitive steps reach the target.

Limits. These are ideal points, not thick trees: an almost-collinear dot does not block. On these rectangular grids every intermediate lattice point is present. Visibility could differ if dots were removed. The source’s asymptotic density discussion is not part of this lesson.

What each group is doing

K-1: Test exact sight lines, find first blockers, and compare rows and rays. Counting to 6 and holding a thread straight suffice; coordinates may be spoken or pointed to. A demonstration of a blocker is a valid explanation of a particular hidden target.

Grades 2-3: Group dots sharing a direction and conjecture a divisibility rule from examples, then explain a fully visible row. Coordinates mean “across, up”; small equal groups and multiplication are helpful.

Grades 4-5: Explain both directions of the visibility criterion and why the blocker list is complete. Division, common factors and proportional steps are the prerequisites. The reduced-fraction argument can stay with the facilitator if children can only justify particular rays.

Materials and preparation

For five pairs: five 30-40 cm pieces of thin thread, removable tape, pencils, straightedges and printed dot grids. On the 0-6 boards, tape the thread down at O rather than holding it with a finger, which can cover (1,1), the near miss beside the sight line to (6,5); avoid pins with young children. Rehearse this once; it is untested. A clear ruler is a useful alternative when thread thickness obscures a dot. One partner holds O, or presses the tape, while the other tests the target, then swap. Print enlarged grids only if hands or counters need more room.

Related prior mathematics: Week 9’s bouncing-path work uses primitive first lattice points, and Week 4 uses gcd in ring steps. Here children investigate exact visibility and blocker counts; it is a different task, not a first-ever appearance of gcd. Unpiloted.

Launch and a flexible hour

0-5, after the opening run-around: Launch. Let children stretch the thread, then use the printed example: O=(0,0), T=(4,2), B=(2,1). Keep O fixed; B lies exactly between O and T, so write B on it. Contrast T=(3,2), which has no intervening dot, and circle it as visible. Ask children to show the difference without introducing gcd yet.

5-20: K P1-2, 2-3 P1-2, upper P1-2. **20-38:** Choose blocker counts, row puzzles or the general rule; a pair can stand and inspect the other pair's claimed sight line. **38 to the end:** Share a prediction and its exact explanation. For younger children, stop with a reliably tested ray or row; a factor rule is not required. Minutes count from the launch; the tables have about 35-40 minutes.

Visible-dot catalog: P1 in every band

Each row gives the across-coordinates x of *all* visible dots at height y. Exclude O. The K grid has 13 visible dots; the 0-6 grid has 25.

Height y	K grid: x=0...4	Larger grid: x=0...6
0	1	1
1	0,1,2,3,4	0,1,2,3,4,5,6
2	1,3	1,3,5
3	1,2,4	1,2,4,5
4	1,3	1,3,5
5	Outside this grid	1,2,3,4,6
6	Outside this grid	1,5

K-1 answer key, P2-5

P2. First picture: targets (4,4),(4,2),(2,4),(3,3) are all hidden; their nearest blockers are (1,1),(2,1),(1,2),(1,1), respectively. Second picture: (4,0) is blocked first by (1,0); (0,4) by (0,1); (2,2) by (1,1); (4,3) is visible and has no blocker.

P3. Exactly one blocker: (0,2),(2,0),(2,2),(2,4),(2,6),(4,2),(4,6),(6,2),(6,4). Exactly two: (0,3),(3,0),(3,3),(3,6),(6,3). Their gcds are 2 and 3 respectively. Use thread to check; children need not write these coordinate lists. An adult can write on the lines, or a child who does not write can mark the two kinds of target in two colours on the grid.

P4. Row y=1 is the only row where every dot is visible, from (0,1) to (6,1). Row 0 has the hidden dot (2,0), and each row y=2,...,6 has the hidden dot (0,y). No row is all hidden, since (1,y) is visible in every row, including (1,0). A child who turns the board may find column x=1, which is also all visible.

P5. Three visible dots tie: (1,0),(0,1),(1,1). Each hides five dots, its multiples 2 through 6. Every other visible dot has some coordinate at least 2, so it reaches at most three dots within this grid and hides at most two.

Grades 2-3 answer key

P1. Use the complete catalog on page 2. Three possible first-blocker marks: (6,4) → (3,2), (6,3) → (2,1), (4,4) → (1,1). Accept any three correct hidden targets.

P2. The requested complete interior lists are below. The target itself never goes in the blocker list.

Target	All blockers between O and target
(6,4)	(3,2)
(6,3)	(2,1), (4,2)
(5,3)	None
(5,2)	None
(4,4)	(1,1), (2,2), (3,3)
(4,1)	None

P3. The two complete target lists are exactly K P3 on page 2, including axis points. “One blocker” means gcd 2, not just “both coordinates even”: (4,4) has three blockers and (6,6) has five.

P4. Excluding O, the five groups on the 0-6 grid are:
(1,1): (1,1),(2,2),(3,3),(4,4),(5,5),(6,6).
(2,1): (2,1),(4,2),(6,3).
(3,2): (3,2),(6,4).
(1,2): (1,2),(2,4),(3,6).
(1,0): (1,0),(2,0),(3,0),(4,0),(5,0),(6,0).
These are all positive multiples of the named first point that fit in the grid.

P5. A dot is visible exactly when the two coordinates have no common divisor greater than 1. Thus (12,8) is hidden (first point (3,2), three blockers); (10,7) is visible; (15,5) is hidden (first (3,1), four blockers); (11,1) is visible. Dividing both coordinates by a common divisor gives a smaller step in the same straight direction. The converse needs the reasoning on page 4.

P6. Row y=1: every dot (x,1), x≥0, is visible. Every interior point of the segment to (x,1) has height strictly between 0 and 1, so cannot be a lattice point. This is an exact geometric proof, not an inference from a finite row. It is the only such row: row 0 has the hidden dot (2,0), and each row y≥2 has the hidden dot (0,y). Accept column x=1 as well.

Hints, only when needed

For a near miss, ask whether the thread passes through the center of the dot. For a first blocker, shorten the thread along the same ray. For a number rule, compare (6,3), (6,4) and (5,3), rather than add a large coordinate chart. For a full blocker list, ask what equal “across/up” step repeats along that ray.

Grades 4-5 answer key and reasoning

P1. The 25 visible dots are in the catalog on page 2. A hidden target has one blocker fewer than the gcd of its coordinates, and on this grid the gcd is at most 6, so the possible counts are 1, 2, 3, 4 and 5. The diagonal shows them all: (2,2) has 1 blocker, (3,3) has 2, (4,4) has 3, (5,5) has 4 and (6,6) has 5. The axes work the same way, and (6,4) or (6,3) give 1 or 2 off the diagonal. Other correct choices are welcome.

P2. For (6,4), first (3,2), only blocker (3,2). For (6,3), first (2,1), blockers (2,1),(4,2). For (5,3), first (5,3) itself, no blockers. For (6,6), first (1,1), blockers (k,k), k=1,...,5. For (6,0), first (1,0), blockers (k,0), k=1,...,5. For (0,6), first (0,1), blockers (0,k), k=1,...,5.

P3. Targets (12,8),(15,10),(21,14) all have first point (3,2) and respectively 3,4,6 blockers. For (25,15), first (5,3), four blockers. For (13,8), first (13,8), zero blockers. The rule applies to every nonzero whole-coordinate target; O itself has no direction and is excluded. The extra grid on that page runs to (25,15) at 7 mm; about 20 cm of thread reaches the farthest. (12,8), (15,10) and (21,14) can be checked with thread; (25,15) is close (the nearest other dot misses by 1.2 mm), and (13,8) cannot be: its thread passes 0.46 mm from the centers of (5,3) and (8,5), so any real thread touches both 0.77 mm dots. That is the moment the rule beats the thread: 5×8 and 3×13 differ by 1.

P4. For (6,4), the group is {(3,2),(6,4)}. For (6,3), it is {(2,1),(4,2),(6,3)}. For (6,6), it is {(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)}. Each nonzero target determines one ray and that ray has one first lattice point, so the groups are disjoint.

P5. If $q>1$ divides a and b, then (a/q,b/q) is a lattice point strictly between O and (a,b). Conversely, a blocker $(u,v)=t(a,b)$ has $0<t<1$. Write $t=r/s$ in lowest terms. Since ra/s and rb/s are integers and $\gcd(r,s)=1$, s divides both a and b; also $s>1$. Thus every blocker certifies a nontrivial common divisor. Do not assume every blocker is the first one or that its coordinates divide the target coordinates.

P6. Put u=a/d, v=b/d; then $\gcd(u,v)=1$. The d-1 points $k(u,v)$, $1\le k<d$, are blockers. Any lattice point on that ray is $t(u,v)$. For a positive coordinate, t is rational; write it in lowest terms. Its denominator must divide both u and v, hence is 1. Therefore t is an integer, so no extra points fit between consecutive listed points. This also covers axes: the primitive coordinate pair is (1,0) or (0,1).

P7. Row y=1 is fully visible by the height argument on page 3. The points (n,n), $n\ge2$, form an infinite hidden collection, all behind (1,1). Target (100,100) has exactly 99 blockers, (k,k) for k=1,...,99; (100,0) is another valid example.

Optional extension and primary source

Move O to (1,1) on a fresh grid and ask which sight lines change. Work with displacement coordinates (x-1,y-1); for example (3,3) remains hidden behind (2,2), while (4,2), hidden from O behind (2,1), is now visible, and (3,5), visible from O, is now hidden behind (2,3). This tests whether the rule describes absolute labels or the viewer’s direction.

Karl-Dieter Crisman, *Number Theory: In Context*, §24.6.1, “Random integer lattice points,” line-of-sight figure and common-divisor explanation. Read the author-written text. Only its visibility model is used; the exact finite-grid lists and converse proof above were independently checked.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

Keep one adult at each table; an adult may hold the origin while the child chooses targets. On return, K-1 can compare blocker counts (P3), rows (P4), or which visible dots hide the most (P5). Grades 2-3 can group dots by direction (P4), then use the number-rule and visible-row questions (P5-6). Grades 4-5 can predict far targets (P3), complete ray groups (P4), explain both directions of the criterion (P5), or justify exact blocker counts (P6-7). Saved sight lines should pass through dot centers; thread thickness is not a mathematical blocker. On the 7 mm grid of Grades 4-5 P3 the thread cannot settle (13,8): it passes 0.46 mm from the centers of (5,3) and (8,5), and there the rule beats the thread.